



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Measure Theory Crash Course

**σ -Algebras, Measures, Outer Measure, Lebesgue Measure, and
Non-Measurable Sets**

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CONDUCTED BY

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Why Measure Theory?

1. σ -Algebras: The Measurable Sets

σ -Algebra

Let X be a set. A **σ -algebra** $\mathcal{A}$ on X is a collection of subsets of X such that:

1. $X \in \mathcal{A}$;
2. if $A \in \mathcal{A}$, then $A^c = X \setminus A \in \mathcal{A}$;
3. if $A_1, A_2, \dots \in \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

The pair $(X, \mathcal{A})$ is called a **measurable space**, and the members of $\mathcal{A}$ are called **measurable sets**.

Basic Examples

For every set X :

1. $\{\emptyset, X\}$ is the **trivial** σ -algebra;
2. $\mathcal{P}(X)$ is the **power-set** σ -algebra;
3. if $X = 1, 2, 3, 4$ and $\mathcal{A} = \{\emptyset, \{1, 2\}, \{3, 4\}, X\}$, then $\mathcal{A}$ is a σ -algebra on X .

💡 Consequences of the Axioms

If $\mathcal{A}$ is a σ -algebra on X , then:



1. $\emptyset \in \mathcal{A}$;
2. $\mathcal{A}$ is closed under finite unions and finite intersections;
3. if $A_1, A_2, \dots \in \mathcal{A}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$;
4. if $A, B \in \mathcal{A}$, then $A \setminus B \in \mathcal{A}$.

💡 Intersection of σ -Algebras

Let $\{\mathcal{A}_i\}_{i \in I}$ be any nonempty collection of σ -algebras on the same set X . Then

$$\bigcap_{i \in I} \mathcal{A}_i$$

is a σ -algebra on X .

Generated σ -Algebra

If $\mathcal{C} \subseteq \mathcal{P}(X)$, the σ -**algebra generated by** $\mathcal{C}$ is

$$\sigma(\mathcal{C}) = \bigcap \{ \mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{C} \subseteq \mathcal{A} \}.$$

Thus $\sigma(\mathcal{C})$ is the smallest σ -algebra containing $\mathcal{C}$.

Borel σ -Algebra

The **Borel σ -algebra** on $\mathbb{R}$ is

$$\mathcal{B}(\mathbb{R}) = \sigma\left(\{(a, b) : a < b, a, b \in \mathbb{R}\}\right).$$

Its elements are called **Borel sets**.

② Cardinality of the Borel σ -Algebra

Show that $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$. Hint: use the countable family of rational-endpoint intervals for the upper bound, and note that all singletons are Borel for the lower bound.

2. Measures and Measure Spaces

Measure

Let $(X, \mathcal{A})$ be a measurable space. A **measure** is a function 

$$\mu : \mathcal{A} \longrightarrow [0, \infty]$$

such that:

1. $\mu(\emptyset) = 0$;
2. if $A_1, A_2, \dots \in \mathcal{A}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The triple $(X, \mathcal{A}, \mu)$ is called a **measure space**.

② Counting Measure

On $(X, \mathcal{P}(X))$, define

$$\#(A) = \begin{cases} \text{the number of elements of } A, & A \text{ is finite,} \\ \infty, & A \text{ is infinite.} \end{cases}$$

Show that $\#$ is a measure. Determine when it is finite and when it is σ -finite.

② Dirac Measure

Let $(X, \mathcal{A})$ be a measurable space and fix $x_0 \in X$. Define

$$\delta_{x_0}(A) = \begin{cases} 1, & x_0 \in A, \\ 0, & x_0 \notin A. \end{cases}$$

Show that δ_{x_0} is a finite measure on $(X, \mathcal{A})$.

② Probability Measure

Let $(X, \mathcal{A})$ be a measurable space. A measure μ is a **probability measure** if $\mu(X) = 1$. Where we assume that $\mu(X) = 1$, we can interpret $\mu(A)$ as the probability of the event $A \in \mathcal{A}$.

② Borel Measure

Let $\mathcal{A} = \mathcal{B}(\mathbb{R})$. There is a unique measure λ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that

$$\lambda((a, b)) = b - a \quad (a < b).$$

This is called **Borel measure**.

Also for half open or closed intervals, $\lambda((a, b]) = \lambda([a, b)) = \lambda([a, b]) = b - a$.

💡 Basic Properties of Measures

Let $(X, \mathcal{A}, \mu)$ be a measure space. Then:

1. **Monotonicity:** if $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
2. **Difference formula:** if $A \subseteq B$ and $\mu(A) < \infty$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

3. **Countable subadditivity:**

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

4. **Continuity from below:** if $A_n \uparrow A$, then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

5. **Continuity from above:** if $A_n \downarrow A$ and $\mu(A_1) < \infty$, then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

3. Outer Measure: Measuring Every Subset

Outer Measure

An **outer measure** on a set X is a function

$$\mu^* : \mathcal{P}(X) \longrightarrow [0, \infty]$$

such that:

1. $\mu^*(\emptyset) = 0$;
2. if $A \subseteq B \subseteq X$, then $\mu^*(A) \leq \mu^*(B)$;
3. for every sequence $A_1, A_2, \dots \subseteq X$,

$$\mu^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n).$$

Unlike a measure, an outer measure is defined on *all* subsets of X , but it need not be additive on all of them.

Length of an Interval

For any bounded interval I with endpoints $a \leq b$, define 

$$\ell(I) = b - a,$$

regardless of whether either endpoint belongs to I . Set $\ell(\emptyset) = 0$.

Lebesgue Outer Measure on $\mathbb{R}$

For every $A \subseteq \mathbb{R}$, define 

$$\lambda^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : A \subseteq \bigcup_{n=1}^{\infty} I_n, I_n \text{ bounded open intervals} \right\}.$$

We use the convention $\inf \emptyset = \infty$.

💡 Properties of Lebesgue Outer Measure

For all $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$:



1. if I is a bounded interval, then $\lambda^*(I) = \ell(I)$;
2. $\lambda^*(A + t) = \lambda^*(A)$, where $A + t = \{x + t : x \in A\}$;
3. if A is countable, then $\lambda^*(A) = 0$.

② Countable Sets Have Outer Measure zero

If $A = \{a_1, a_2, \dots\}$ and $\varepsilon > 0$, cover a_n by an interval of length $\varepsilon/2^n$.
Use this to prove $\lambda^*(A) = 0$.

② Outer Measure Need Not Be Additive

Explain why we cannot yet call λ^* a measure on $\mathcal{P}(\mathbb{R})$. Which axiom is missing?

4. Carathéodory Measurability

Carathéodory Criterion

Let μ^* be an outer measure on X . A set $E \subseteq X$ is μ^* -**measurable** if, for every $A \subseteq X$,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \setminus E).$$

Thus a measurable set splits every test set A without creating or losing outer measure.

💡 Carathéodory's Theorem

Let μ^* be an outer measure on X , and let

$$\mathcal{M}_{\mu^*} = \{E \subseteq X : E \text{ is } \mu^* \text{-measurable}\}.$$

Then $\mathcal{M}_{\mu^*}$ is a σ -algebra, and the restriction $\mu^*|_{\mathcal{M}_{\mu^*}}$ is a complete measure.



5. Lebesgue Measure

Lebesgue Measurable Sets and Lebesgue Measure

Apply Carathéodory's construction to Lebesgue outer measure λ^* . The resulting σ -algebra is denoted by $\mathcal{L}(\mathbb{R})$ and its members are the **Lebesgue measurable sets**. The **Lebesgue measure** is

$$\lambda = \lambda^*|_{\mathcal{L}(\mathbb{R})}.$$

Hence, for $E \in \mathcal{L}(\mathbb{R})$,

$$\lambda(E) = \lambda^*(E).$$

💡 Borel Sets Versus Lebesgue Measurable Sets

Every Borel set is Lebesgue measurable, and

$$\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L}(\mathbb{R}) \subsetneq \mathcal{P}(\mathbb{R}).$$

The restriction $\lambda|_{\mathcal{B}(\mathbb{R})}$ is called **Borel Lebesgue measure**. It is the unique measure on $\mathcal{B}(\mathbb{R})$ satisfying

$$\lambda((a, b]) = b - a \quad (a < b).$$

Complete Measure Space

A measure space $(X, \mathcal{A}, \mu)$ is **complete** if whenever $N \in \mathcal{A}$, $\mu(N) = 0$, and $S \subseteq N$, then $S \in \mathcal{A}$ (and necessarily $\mu(S) = 0$).

Completion of Borel Lebesgue Measure

The Lebesgue σ -algebra is the completion of the Borel σ -algebra with respect to Borel Lebesgue measure. Equivalently, $E \subseteq \mathbb{R}$ is Lebesgue measurable if and only if there exist a Borel set B and a subset N of a Borel null set such that

$$E = B \cup N.$$

💡 Principal Properties of Lebesgue Measure

Lebesgue measure is complete, translation invariant, and countably additive. In particular, if $E \in \mathcal{L}(\mathbb{R})$ and $t \in \mathbb{R}$, then $E + t$ is measurable and

$$\lambda(E + t) = \lambda(E).$$

Null Set

A Lebesgue measurable set $N \subseteq \mathbb{R}$ is a **null set** if $\lambda(N) = 0$. 

Countable Sets Are Null

Every countable subset of $\mathbb{R}$ is Lebesgue measurable and has measure zero. In particular, 

$$\lambda(\{x\}) = 0 \quad \text{and} \quad \lambda(\mathbb{Q}) = 0.$$

6. An Uncountable Null Set: The Cantor Set

Cantor Set

Start with $C_0 = [0, 1]$. From every interval in C_{n-1} remove its open middle third, obtaining C_n . The **Cantor set** is

$$C = \bigcap_{n=0}^{\infty} C_n.$$

At stage n , the set C_n is a union of 2^n closed intervals, each of length 3^{-n} .

💡 Basic Properties of the Cantor Set

The Cantor set C is closed, uncountable, has empty interior, and satisfies

$$\lambda(C) = 0.$$

❓ Why Is the Cantor Set Null (Measure of zero)?

Since $C \subseteq C_n$, use

$$\lambda(C_n) = 2^n 3^{-n} = \left(\frac{2}{3}\right)^n$$

and let $n \rightarrow \infty$.

② Why Is the Cantor Set Uncountable?

Show that $x \in C$ precisely when x has a ternary expansion using only the digits 0 and 2 (with the usual care at endpoints). Relate such expansions to infinite binary sequences.

② Cardinality of the Lebesgue σ -Algebra

Prove that

$$|\mathcal{L}(\mathbb{R})| = 2^{\mathfrak{c}}.$$

Hint: $|C| = \mathfrak{c}$ and every subset of the null set C is Lebesgue measurable.

Comparison of Borel and Lebesgue Measurability

💡 Strictness of the Borel–Lebesgue Inclusion

Since $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$ but $|\mathcal{L}(\mathbb{R})| = 2^{\mathfrak{c}}$, there exist Lebesgue measurable sets that are not Borel sets.

7. Almost Everywhere

Almost Everywhere

A property $P(x)$ holds **almost everywhere** (a.e.) on a measure space $(X, \mathcal{A}, \mu)$ if there exists a measurable null set N such that $P(x)$ holds for every $x \in X \setminus N$.

If the failure set $\{x \in X : P(x) \text{ fails}\}$ is measurable, this is equivalent to saying that the failure set has measure zero.

② Equality Almost Everywhere

Let $f, g : X \rightarrow \mathbb{R}$. Write precisely what $f = g$ a.e. means. Show that “equal almost everywhere” is an equivalence relation (assuming the relevant sets are measurable).

8. Not Every Subset Is Lebesgue Measurable

Vitali Equivalence Relation

On $[0, 1]$, define

$$x \sim y \iff x - y \in \mathbb{Q}.$$

Using the axiom of choice, select one representative from each equivalence class. Any resulting set $V \subseteq [0, 1]$ is called a **Vitali set**.

A Vitali Set Is Not Lebesgue Measurable

A Vitali set V is not Lebesgue measurable.

❓ Vitali Argument

Let $\mathbb{Q} \cap [-1, 1] = \{q_1, q_2, \dots\}$. Show that the sets $V + q_n$ are pairwise disjoint, and that

$$[0, 1] \subseteq \bigcup_{n=1}^{\infty} (V + q_n) \subseteq [-1, 2].$$

Assuming V were measurable, use translation invariance and countable additivity to obtain a contradiction, considering separately the cases $\lambda(V) = 0$ and $\lambda(V) > 0$.

9. Why Countable Additivity?

② Finite Additivity Is Too Weak

Countable additivity implies finite additivity, but not conversely. Explain why finite additivity alone does not force

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

for disjoint sets A_n , and why this would break continuity from below.

② Why Not Uncountable Additivity?

Every interval is the uncountable disjoint union of its singleton points. If a length measure were additive over arbitrary disjoint families, then

$$\lambda([0, 1]) = \sum_{x \in [0, 1]} \lambda(\{x\}) = 0,$$

contradicting $\lambda([0, 1]) = 1$. Explain why countable additivity avoids this contradiction.

10. Higher-Dimensional Lebesgue Measure

Volume of a Rectangle in $\mathbb{R}^n$

For a half-open rectangle

$$R = \prod_{j=1}^n (a_j, b_j], \quad \text{vol}(R) = \prod_{j=1}^n (b_j - a_j).$$



Lebesgue Outer Measure on $\mathbb{R}^n$

For a half-open rectangle

$$R = \prod_{j=1}^n (a_j, b_j], \quad \text{vol}(R) = \prod_{j=1}^n (b_j - a_j).$$

For $A \subseteq \mathbb{R}^n$, define

$$\lambda_n^*(A) = \inf \left\{ \sum_{k=1}^{\infty} \text{vol}(R_k) : A \subseteq \bigcup_{k=1}^{\infty} R_k \right\}.$$

Carathéodory's criterion then produces the Lebesgue measurable subsets of $\mathbb{R}^n$ and Lebesgue measure λ_n .

Translation Invariance

If $E \subseteq \mathbb{R}^n$ is Lebesgue measurable and $t \in \mathbb{R}^n$, then $E+t$ is measurable and

$$\lambda_n(E + t) = \lambda_n(E).$$

💡 What the Non-Measurability Results Actually Say

- It is not possible to define the Lebesgue measure on *all* subsets of $\mathbb{R}^n$ while preserving the usual geometric properties of measure.
- **Hausdorff's result (1914):** For every dimension $n \geq 1$, there is no countably additive measure defined on all subsets of $\mathbb{R}^n$ that
 - is invariant under isometries, such as translations and rotations, and
 - assigns measure 1 to the unit cube.
- Moreover, Hausdorff showed that for $n \geq 3$, no such *finitely additive* measure exists.
- **Banach–Tarski paradox (1924):** If $n \geq 3$, a ball in $\mathbb{R}^n$ can be decomposed into finitely many pieces and, using only isometries, these pieces can be reassembled into a ball of any prescribed volume.
- Informally, this means that one could, in principle, decompose a small ball and reassemble the pieces into a much larger one—for example, “reassemble a pea into the sun.”
- The construction of such pieces depends essentially on the *axiom of choice*.
- **Banach's result (1923):** For $n = 1$ and $n = 2$, there do exist finitely additive, isometry-invariant extensions of Lebesgue measure to all subsets of $\mathbb{R}^n$.
- However, these extensions are *not countably additive*.
- In dimensions $n \geq 3$, the obstruction is even stronger: finite additivity cannot be preserved under these requirements.

Summary: The Logical Flow

1. A σ -algebra specifies the sets on which countable set operations are safe.
2. A measure is countably additive on a σ -algebra.
3. Lebesgue outer measure assigns a preliminary size to every subset of $\mathbb{R}$.
4. Carathéodory's criterion selects the sets on which that outer measure behaves additively.
5. Restricting λ^* to those sets gives complete Lebesgue measure.
6. Every Borel set is Lebesgue measurable, but some Lebesgue measurable sets are not Borel.
7. Null sets may be uncountable, while some subsets of $\mathbb{R}$ are not Lebesgue measurable at all.

❓ Concept Check

Explain the differences among the following objects:

$$\mathcal{B}(\mathbb{R}), \quad \mathcal{L}(\mathbb{R}), \quad \mathcal{P}(\mathbb{R}), \quad \lambda^*, \quad \lambda.$$

State the domain of each set function and the inclusion relations among the three collections of sets.

Thank You!

We'd love your questions and feedback.

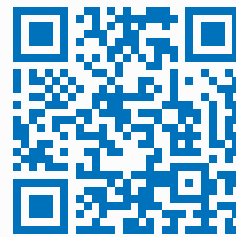
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(Lectures, walkthroughs, and course updates)



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